

Queue Solutions Pty Ltd

AMMBER User Manual

A Motivational Model Builder for Essential Requirements

(AMMBER Version **2025.2**)

April 28, 2026

Table of Contents

1 Introduction 1

1.1 About Motivational Models 1
1.2 Benefits of Motivational Modelling 1
1.3 What is AMMBER? 2

2 Getting Started 3

2.1 How to access AMMBER 3
2.2 Navigating the Interface 3
2.2.1 Landing Page (Home) 3
2.2.2 Editor – Enter Goals / Arrange Hierarchy 5
2.2.3 Editor – Arrange Hierarchy / Render Model 8

3 Core Features 12

3.1 Creating a new model (with default goals) 12
3.2 Editing the goals 13
3.2.1 Adding a new goal 13
3.2.2 Deleting existing goal(s) 14
3.3 Editing the model 18
3.3.1 Changing the colour of a goal in the model 18
3.3.2 Changing the font size within a goal in the model 19
3.4 Exporting the model 20
3.5 Saving the model 20
3.6 Opening a saved model 21
3.7 Staff Mode Instruction 22
3.7.1 Adding feedback commets to model 23
3.7.2 Grading the model 24
3.8 Student Reviewing Grade and Feedback 27
3.9 Model Version control 28
3.9.1 Saving the current model version 29

Bibliography 31

CHAPTER 1
Introduction

1.1 About Motivational Models

Motivational models are diagrams that depict the goals and motivations of a system or a service. They present a hierarchical structure of the goals of the system at a high-level of abstraction. The models capture roles of system stakeholders, the functional goals of the system - what the system will do, the quality goals of the system - how the system will be, and emotional goals, which represent both the desired positive feelings of people when interacting with the system, and possible concerns.

Motivational models have evolved from goal models as presented in agent-oriented modelling [Sterling and Taveter(2009)]. The models have their roots in goal models from the early agent-oriented methodologies, notably Gaia (Wooldridge et al., 2000). ROADMAP as described in (Juan et al, 2002) extended Gaia and was part of significant research in agent-oriented software engineering. In agent-oriented methodologies, goals were used to specify what behaviours agents needed to achieve, though not how the behaviour operated. As described in [Sterling and Taveter(2009)], goals become increasingly more abstract and needed to be depicted at a high level of abstraction.

The current depiction views them as separate from any agent implementation. The notation for motivational models came from [Sterling and Taveter(2009)] and was improved by [Marshall(2018)]. The particular version of the icons drawn by the tool described in this document were designed and implemented by James Marshall.

1.2 Benefits of Motivational Modelling

The models benefit your organisation by having a conversation starter which reflects the organisational or project purpose and tolerates ambiguity. The model can be used later to help clarify purpose and resolve later conflicts in conversation. The models represent a resource a resource to be referred to throughout the lifecycle of a project and can be useful for explaining to clients, and for introducing a project to new members of staff.

Enhancing Communication

A motivational model facilitates communication between diverse stakeholders. Our high-level models are easy to understand and have been tested across hundreds of projects, ensuring clarity and effectiveness.

Allowing for Ambiguity

At the outset of a project, it can be difficult for stakeholders to fully understand their differences. Allowing ambiguity during initial discussions, with later resolution, supports positive collaboration and helps participants align more effectively over time.

Encouraging Emotional Engagement

A key factor in project success is ensuring participants are emotionally engaged. Being explicit about emotional expectations from the start encourages positive participation and strengthens commitment from all involved.

1.3 What is AMMBER?

AMMBER is a tool to help you create motivational models. The acronym is A Motivational Model Builder for Essential Requirements. AMMBER is a refactored and enhanced version of a motivational model editor which was originally developed by a student software engineering team at the University of Melbourne in 2018. Details of the development history preceding AMMBER can be found in MMDEVELOPMENT-link.

CHAPTER 2
Getting Started

2.1 How to access AMMBER

Everyone is welcome to use AMMBER by accessing the AMMBER website at https://motivationalmodelling.github.io/mm-local-editor.

2.2 Navigating the Interface

This section provides an overview of the three main pages of AMMBER. Each page has a specific purpose and set of controls that guide you through building/opening, editing, and visualising motivational models.

2.2.1 Landing Page (Home)

The entry point where you can create a new model or open an existing one.

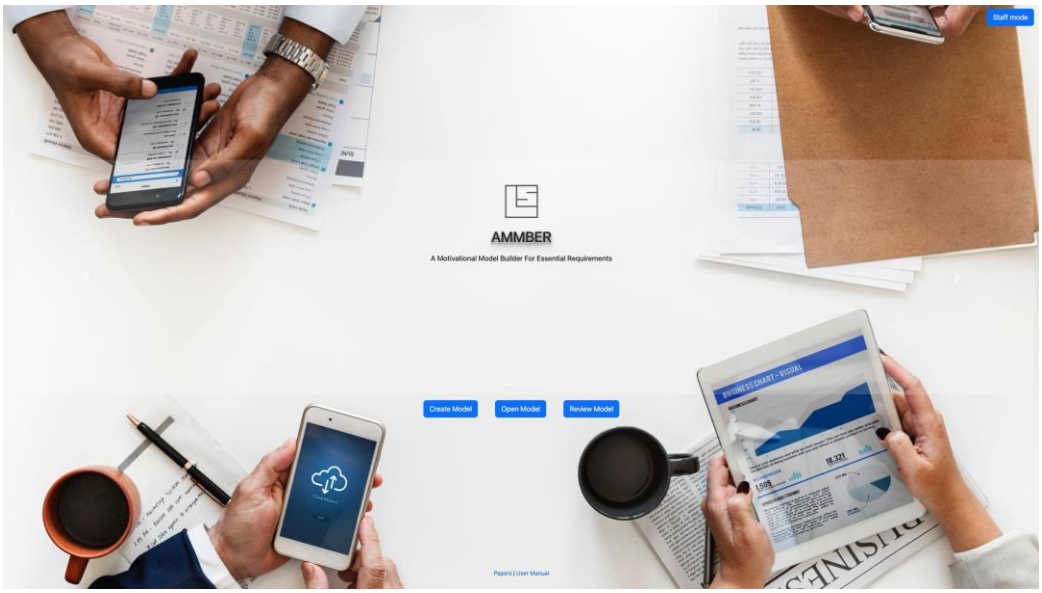


Figure 2.1. AMMBER Home Page.

Main elements:

- Create Model – Creates a new motivational model. If this is your first time using the builder, a default model will be loaded. Clicking this button navigates to the

Editor – Enter Goals / Arrange Hierarchy page.

- Open Model – Allows you to open a model that was created and saved using this builder, stored on your device in JSON format.
- Staff mode – Switch to Staff Mode for lecturer/client to reviewing and grading the model.
- Review Model – Switch to Staff Mode for lecturer/client to reviewing and grading the model.

2.2.1.1. Staff Home Page

When a user clicks the "Staff Mode" button, the system displays a "Name Input Page," requiring the staff member (instructor or client) to enter their name before accessing Staff Mode to view and rate the model.

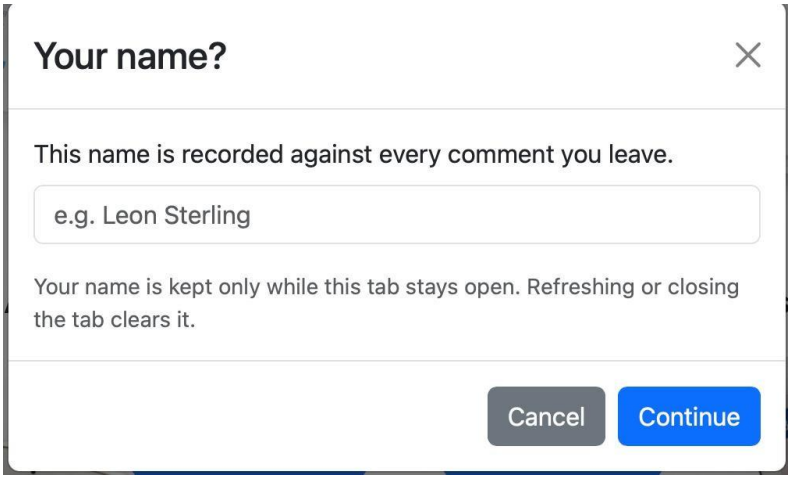


Figure 2.2. Name Confirmation Pop-up

Once the name is submitted, the user is taken to the Staff Mode homepage, shown below.

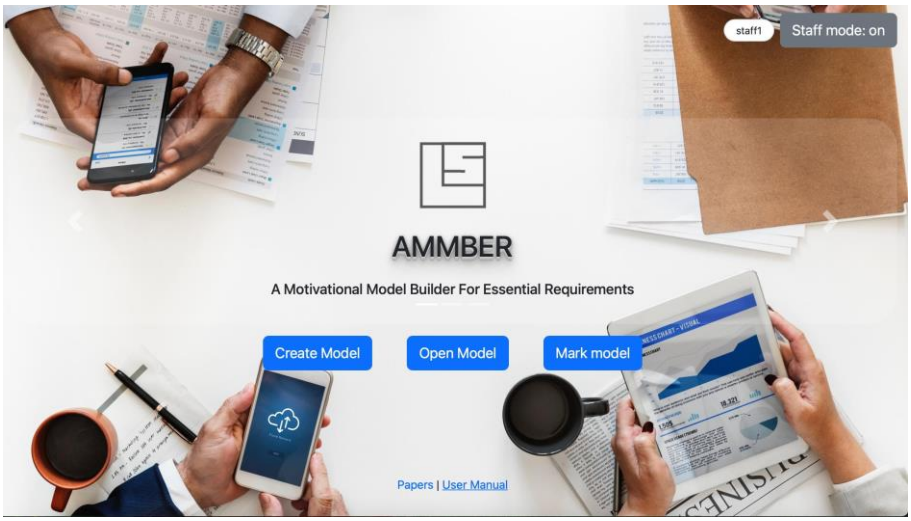
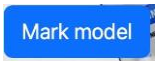
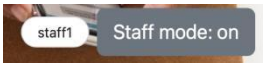


Figure 2.3. Ammber Home Page(Staff Mode)

Main elements:

- 

– Same as in the standard homepage. See 2.2.1 Landing Page (Home) for details.
- 

– Allows staff to open and grade an existing model. Clicking this button navigates to the marking/review interface, where staff can view the model and assign scores or feedback.
- 

– Once Staff Mode is activated, the staffmode button turns grey and displays "Staff mode: on", with the staff member's entered name shown beside it, confirming the current staff identity.

2.2.2 Editor – Enter Goals / Arrange Hierarchy

This page is to enter goals, classify them by type, and arrange them into a hierarchy as the basis of your motivational model.

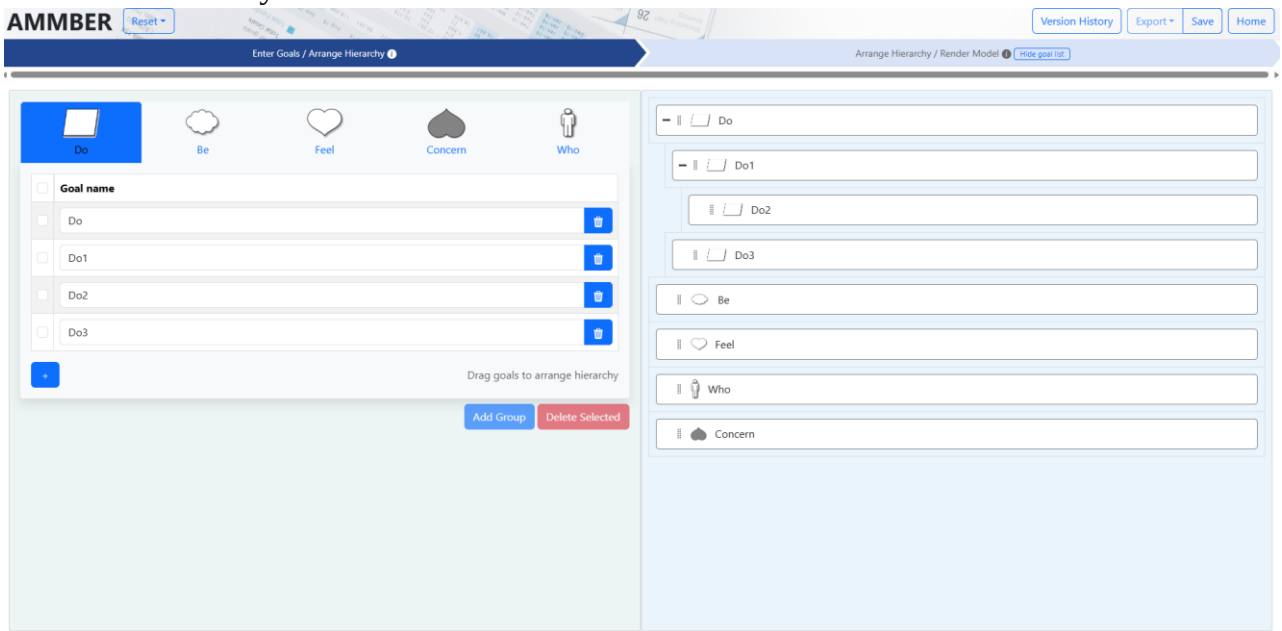


Figure 2.4. Editor – Enter Goals / Arrange Hierarchy Page.

2.2.2.1 Header

There is a header included 4 buttons on top of each page, where the user could click to do a specific task.



Figure 2.5. AMMBER Header.

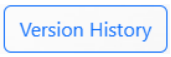
Buttons in the Header:

- 

– Resets the model to either the default model or an empty model.
- 

– Allows you to export the graph in PNG or SVG format. Only available in in Model Builder – Arrange Hierarchy / Render Model page.
- 

– Allows you to save the model in JSON format for future use.
- 

– Navigates back to AMMBER home page.
- 

– Records the editing history of model(both automatically and manually).

2.2.2.2 Navigation Bar

Below the header, there is a navigation bar with two buttons that take the user to the corresponding subpages. Note that if a button is blue, it indicates the current page.



Figure 2.6. Navigation bar for builder subpages.

Main elements in the Navigation Bar:

- 

– Navigate to Editor – Enter Goals / Arrange Hierarchy page.
- 

– Navigate to Editor – Arrange Hierarchy / Render Model page. Clicking this button creates a model based on the goals and hierarchy defined in the hierarchy view.
- 

– Click to show/hide the goal list.

> **Note**
>
> The model is only created after the first time you click this button and only the goals in the hierarchy view will be included (goals in the goal list won't be included).

2.2.2.3 Goal List Panel (Left)

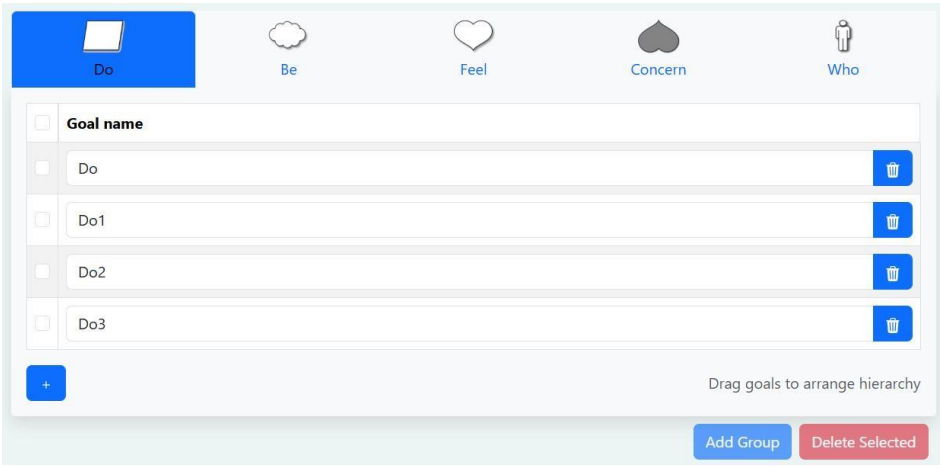
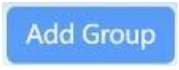


Figure 2.7. Goal List Panel.

In the goal list panel, there are five categories into which items can be entered: do, be, feel, concern, and who. Do, be, feel are goals, who is a stakeholder, and concern is effectively an 'anti-goal,' which correspond to the elements of the do–be–feel framework.

Entering items into these lists is straightforward. Goals can be dragged from the goal list on the left and dropped into the hierarchy view to include them in the model and manage the hierarchy.

Main elements in the Goal List Panel:

-  – Adds a new goal to the goal list.
-  – Deletes an existing goal in the goal list.
-  – Allows you to add a group of selected goals to the hierarchy.
-  – Allows you to delete a group of selected goals from the goal list.

2.2.2.4 Hierarchy Panel (Right)

Goals from the goal list can be transferred into the hierarchy view to support the organisation of hierarchical relationships. Within this view, the relative positions of goals inside a cluster can be adjusted to refine the hierarchical structure. For instance, repositioning a goal horizontally within a cluster enables the construction or modification of parent–child relationships.

To incorporate goals into the hierarchy, you may drag a goal from the goal list and drop it into a cluster. Alternatively, one or more goals can be selected, and the Add Group function can be used to add them simultaneously.

To organise the graph hierarchy, you can drag the goals to different positions. Adjust their positions within a cluster (left or right) to refine the hierarchical structure. Drag left to deepen the hierarchy and right to bring it up a level. You can also hide/expand the sub-goals by clicking on the –/+ sign.

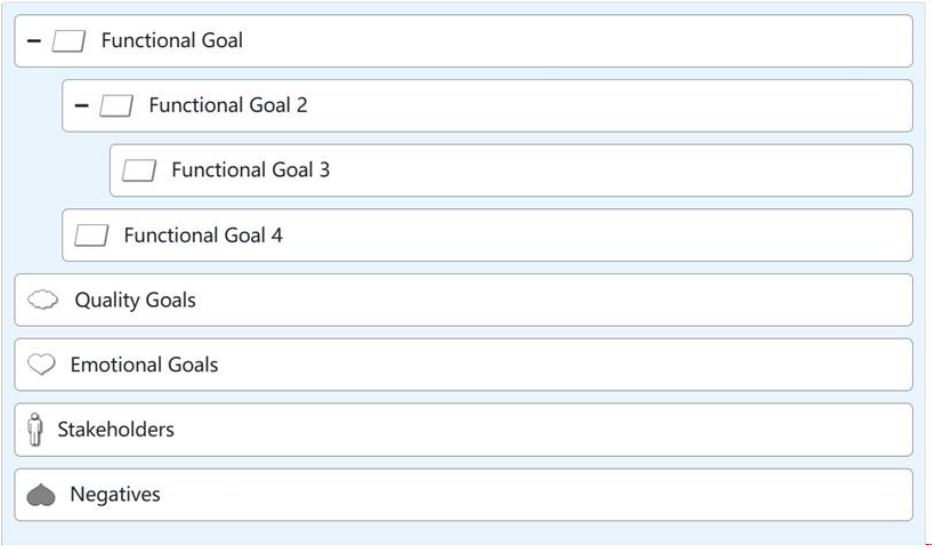


Figure 2.8. Hierarchy Panel.

2.2.3 Editor – Arrange Hierarchy / Render Model

View your motivational model as a diagram for analysis or presentation.

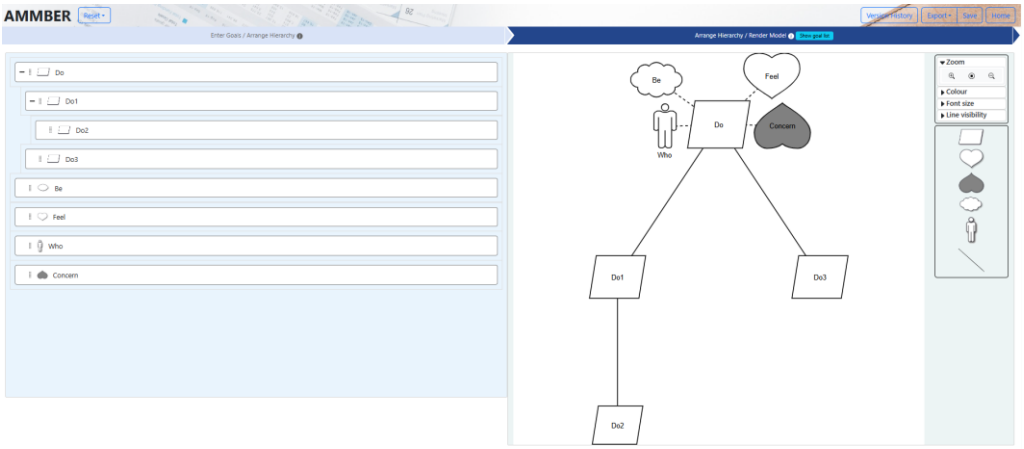


Figure 2.9. Model Builder – Arrange Hierarchy / Render Model Page.

2.2.3.1 Header

See subsubsection 2.2.2.1 for more information.

2.2.3.2 Navigation Bar

See subsubsection 2.2.2.2 for more information.

2.2.3.3 Hierarchy Panel (Left)

See subsubsection 2.2.2.4 for more information.

2.2.3.4 Model Panel (Right)

In the model panel, goals are represented as a structured hierarchical model. The layout of the model is generated automatically to enhance efficiency and reduce manual effort. Nonetheless, users retain the ability to select and drag individual goals to manually adjust their positions when finer control is required.

A toolbar positioned adjacent to the model provides controls for adjusting the model view and modifying goal presentation styles. Additionally, new goals can be introduced directly into the model by dragging the goal symbol from the toolbar into the hierarchy view.

Once the model has been finalised, AMMBER offers functionality to export the model in PNG or SVG formats, as well as to save the model as a JSON file to support future editing or collaborative use.

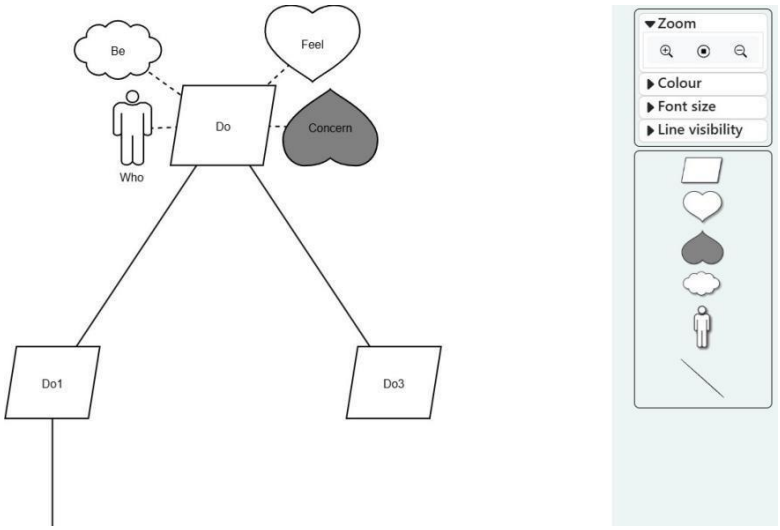
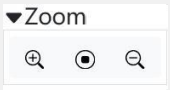


Figure 2.10. Model Panel.

Note

- To change the color or font size, you must first select goal(s). See here for a more detailed guide.
- Use the Zoom Toolbar 

to adjust the view of your model. You can zoom in to see details, fit the model to the screen for an overview, or zoom out to see the overall structure.

2.2.3.5 Feedback Panel (Staff Review)

When AMMBER is opened in staff review mode, a **Feedback panel** appear on the right-hand side of the editor, alongside the goal list or hierarchy view. It remains visible across both the "Enter Goals / Arrange Hierarchy" and "Arrange Hierarchy / Render Model" stages, allowing staff to leave comments on a student's model at any point while reviewing it.

The panel header shows who is currently reviewing the model (e.g. "Reviewing as staff1"). If no comments have been added yet, a placeholder message "No feedback yet. Use + to add the first comment." is displayed.

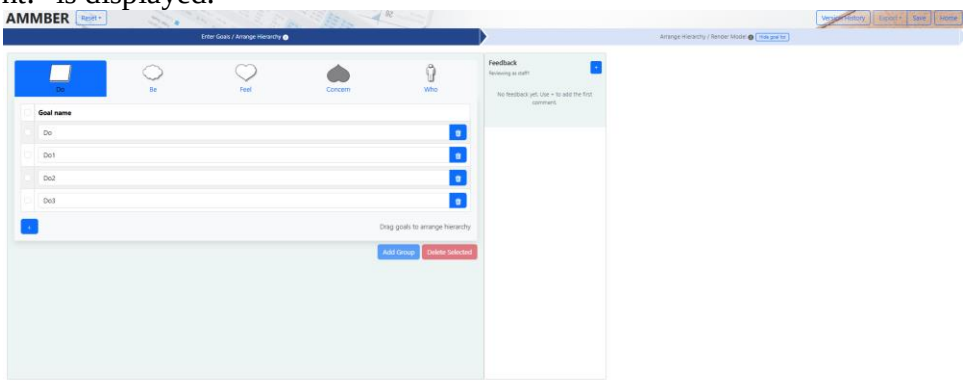


Figure 2.11. Enter Goals / Arrange Hierarchy page with Feedback Panel.

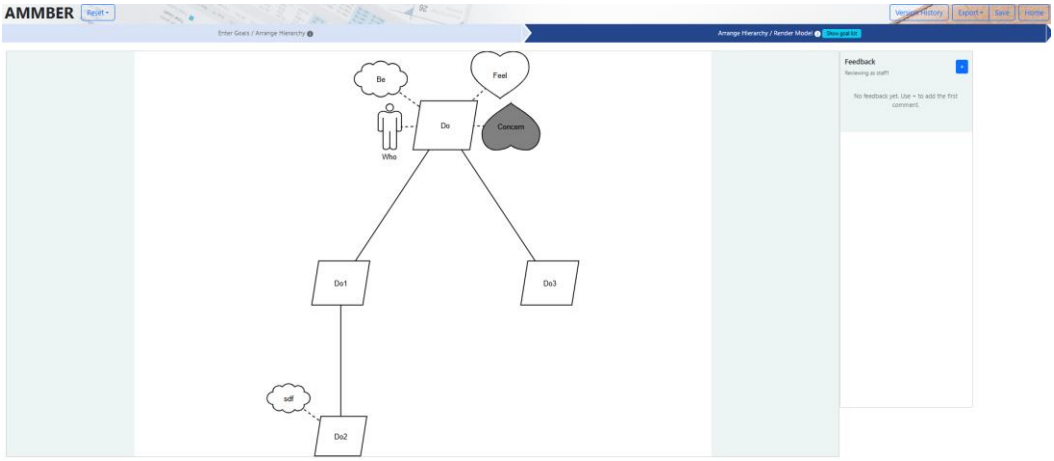


Figure 2.12. Arrange Hierarchy / Render Model page with Feedback Panel.

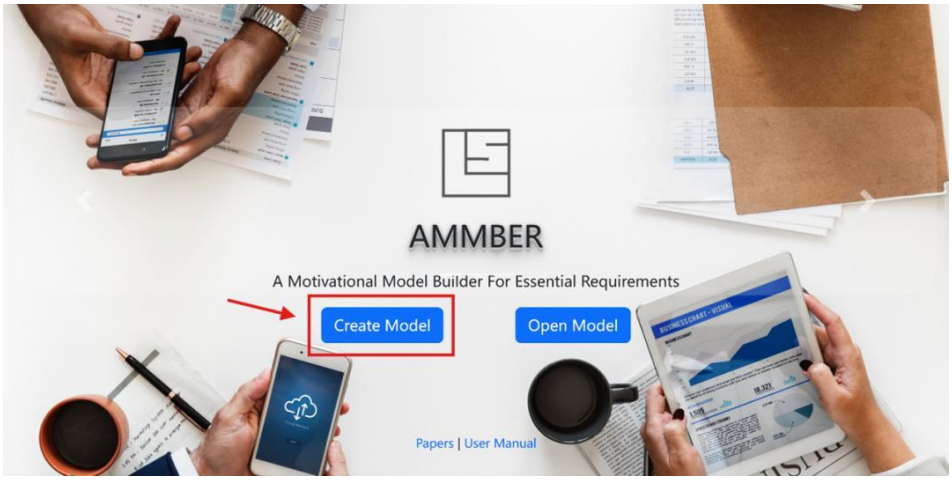
New elements in the Feedback Panel:

-  – Adds a new feedback comment to the panel.

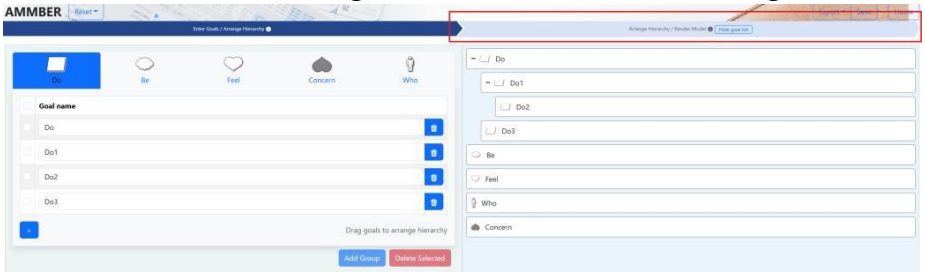
CHAPTER 3
Core Features

3.1 Creating a new model (with default goals)

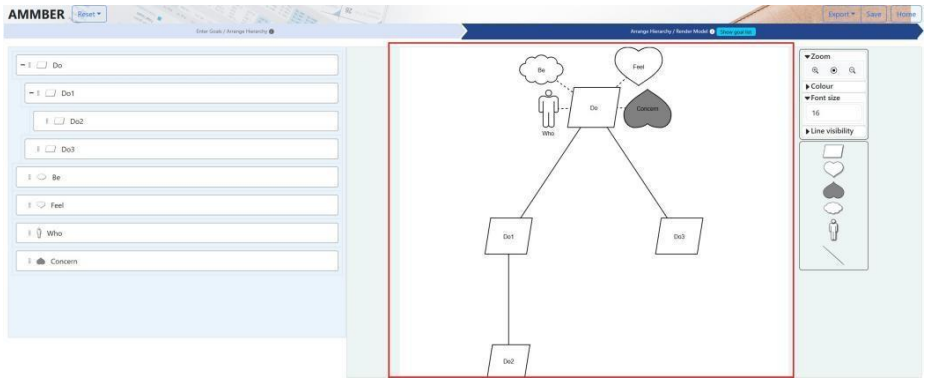
(1) Click the 'Create Model' button in the landing page.



(2) The default goals should appear as shown below. Click "Arrange Hierarchy / Render Model" button to generate the model, or edit the goals if needed.



(3) You can now see the rendered model on the right.



3.2 Editing the goals

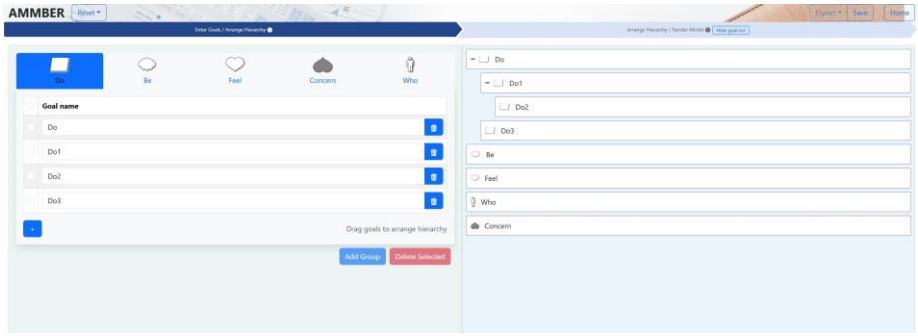
3.2.1 Adding a new goal

You can add a new goal to the model via the goal list or the model toolbar directly.

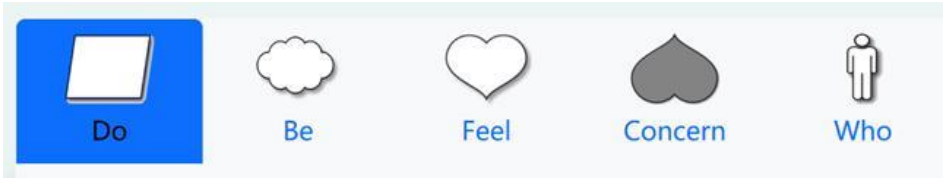
3.2.1.1 Adding a new goal via goal list

There are two main steps to add a new goal to the model via the goal list: first, add the goal to the goal list; then, add the goal to the hierarchy.

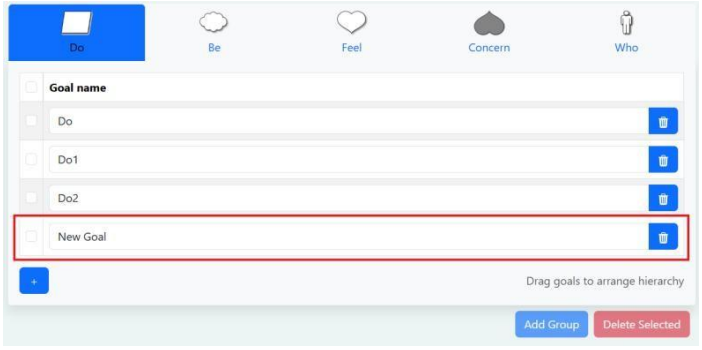
(1) If you have not already, go to the "Enter Goals / Arrange Hierarchy" page.



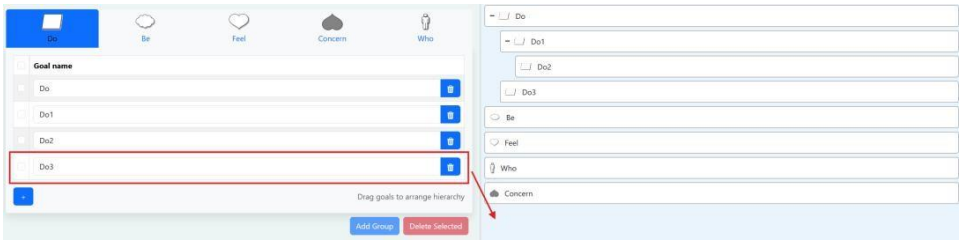
(2) Choose the category of the new goal you want to create.



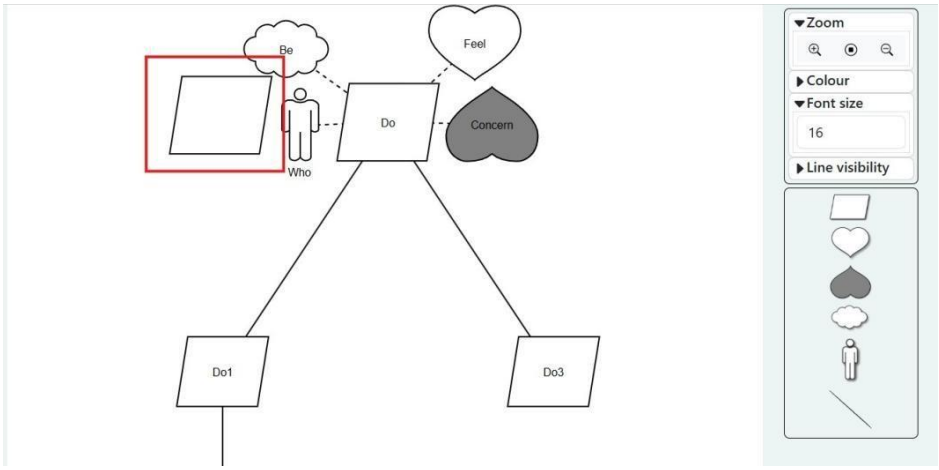
(3) Add a new goal to the goal list by clicking + and entering the goal name.



(4) Drag the goal to the hierarchy panel on the left.

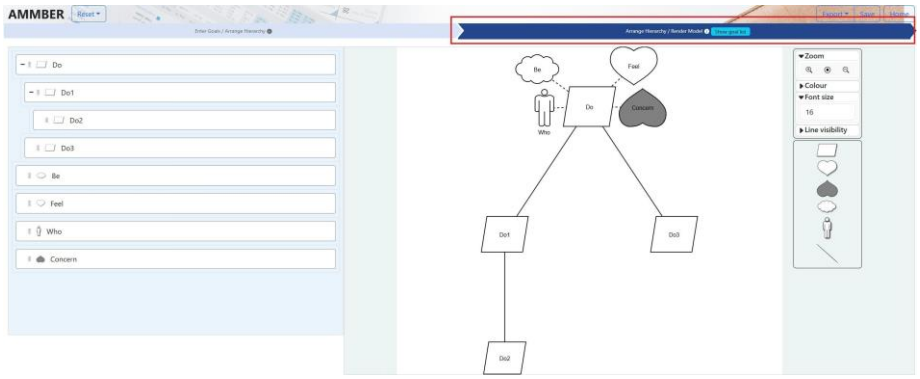


(5) The goal will now appear in the rendered model if you go to the model panel.

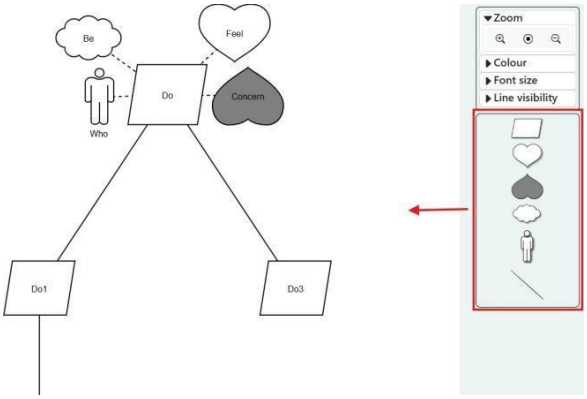


3.2.1.2 Adding a new goal via model toolbar

(1) If you have not already, go to the "Arrange Hierarchy / Render Model" page.



(2) Drag a goal from the toolbar to the model.



3.2.2 Deleting existing goal(s)

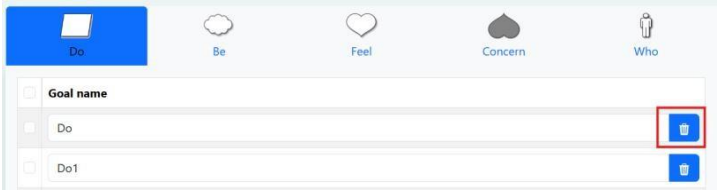
You can delete existing goals individually or as a group. Similar to adding goals, the deletion can be done by multiple ways: via the goal list, the hierarchy view or the rendered model.

Note

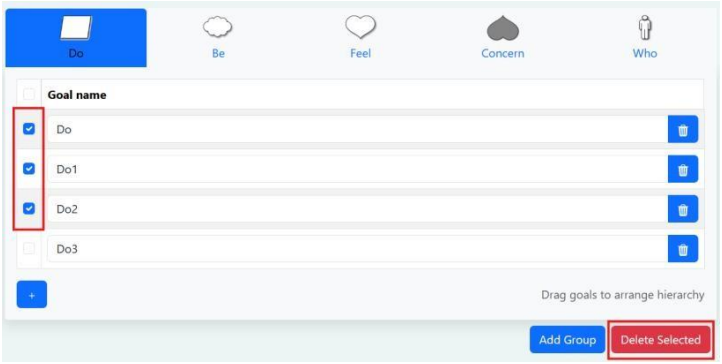
Deleting a goal also deletes its child goals and any copies of it.

3.2.2.1 Deleting existing goal(s) via goal list

◇ To delete a single goal from the list: click  in its row.

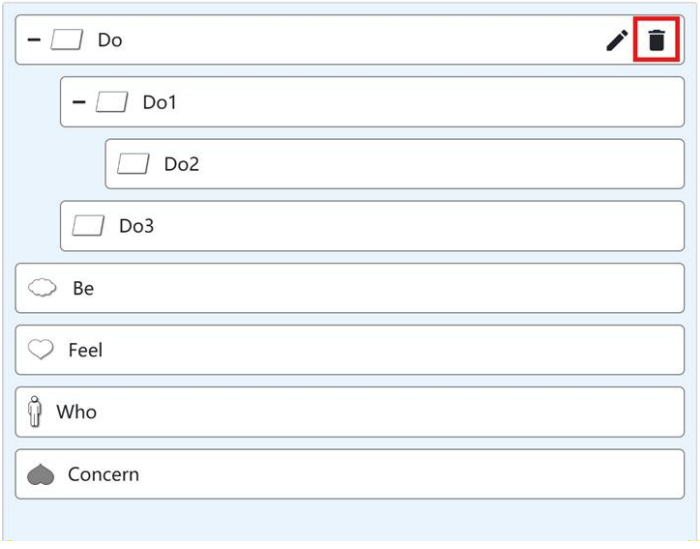


◇ To delete goals as a group: select the goals you want to delete and click  at the bottom of the goal list. Note that you can only delete selected goal groups from one category at a time.

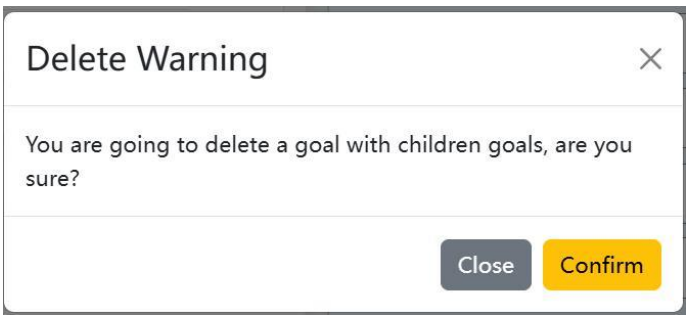


3.2.2.2 Deleting existing goal(s) via hierarchy

(1) In the hierarchy panel, hover over a goal to reveal its action buttons. Click the delete button to remove it.

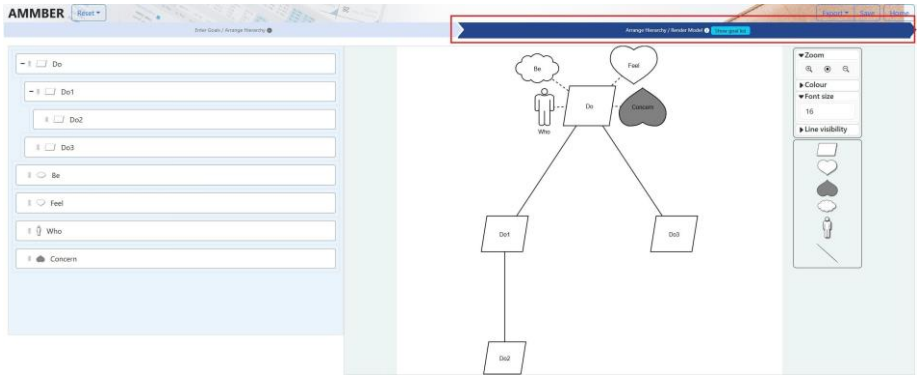


(2) A warning window will pop up if you are trying to delete a goal with child goals. Click Confirm to perform the deletion.

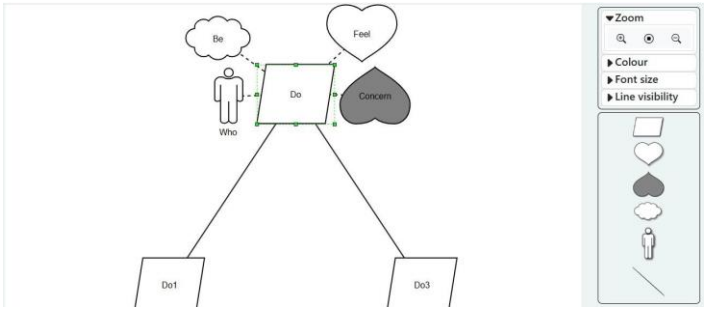


3.2.2.3 Deleting existing goal(s) via rendered model

(1) If you have not already, go to the "Arrange Hierarchy / Render Model" page.



(2) Select goals in the model by clicking on them or dragging a selection box. To select multiple individual goals, hold down the **Ctrl** key (or **Cmd** on Mac) and click on them.

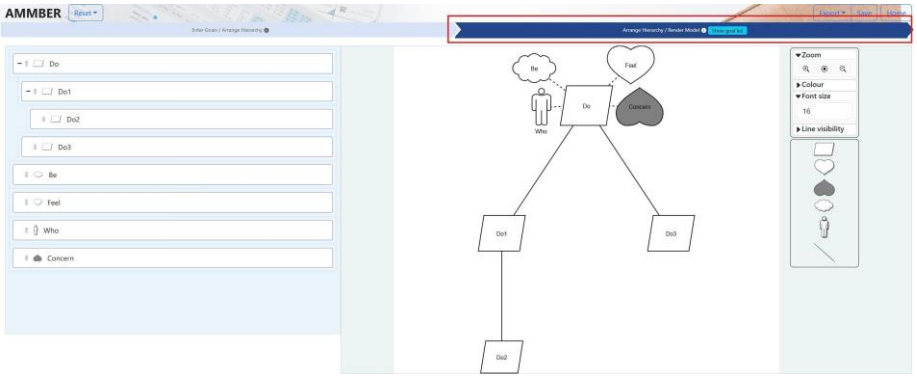


(3) Press the **Delete** key on your keyboard to remove them.

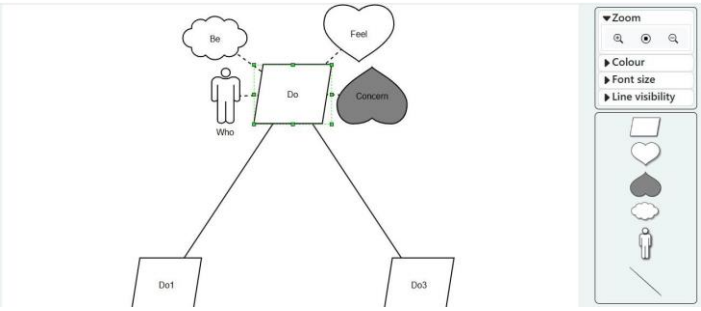
3.3 Editing the model

3.3.1 Changing the colour of a goal in the model

(1) If you haven't already, go to "Arrange Hierarchy / Render Model" page.

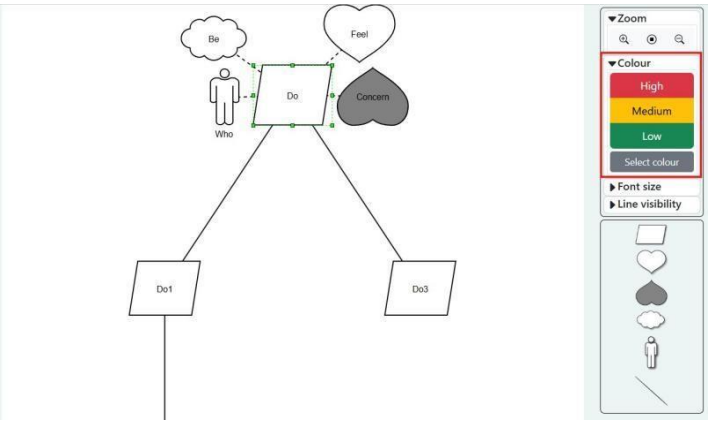


(2) Select goal(s) in the model by clicking on them or dragging a selection box. To select multiple individual goals, hold down the **Ctrl** key (or **Cmd** on Mac) and click on them.

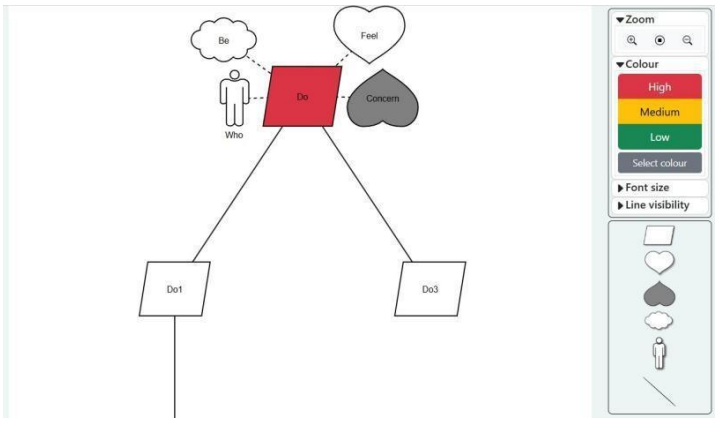


(3) Click ▸Colour in the toolbar to explore the colour options.

(4) Pick a colour option.

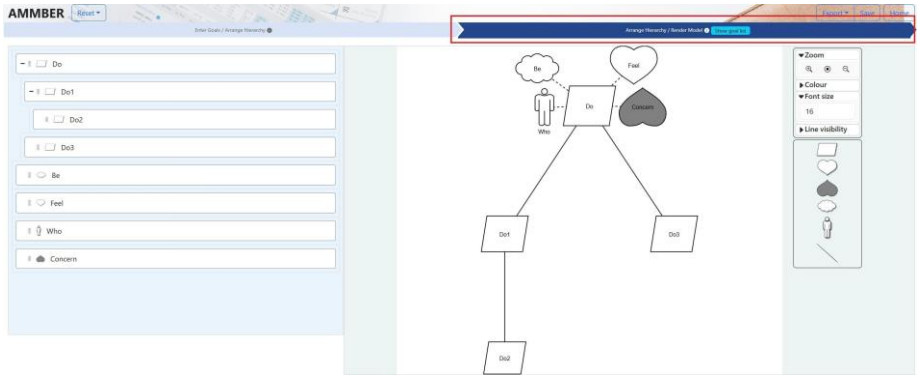


(5) Changes will be applied to the selected goal(s).

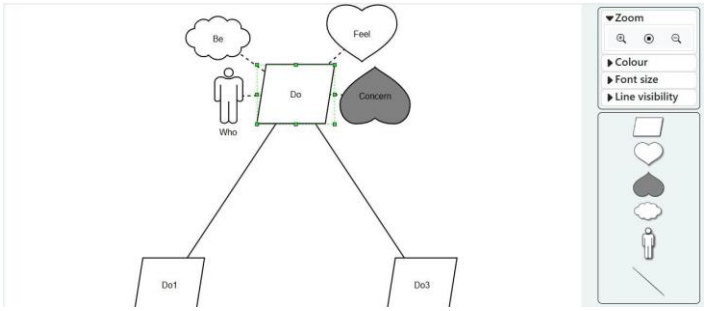


3.3.2 Changing the font size within a goal in the model

(1) If you haven't already, go to "Arrange Hierarchy / Render Model" page.

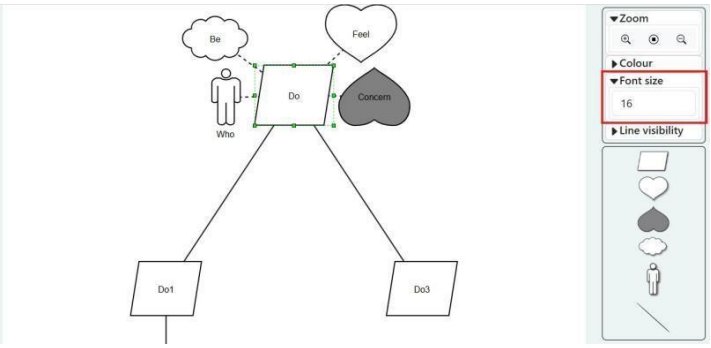


(2) Select goal(s) in the model by clicking on them or dragging a selection box. To select multiple individual goals, hold down the **Ctrl** key (or **Cmd** on Mac) and click on them.



(3) Click ▸Font size in the toolbar to reveal the font controller.

(4) Enter a new font size for the goal(s) you selected.

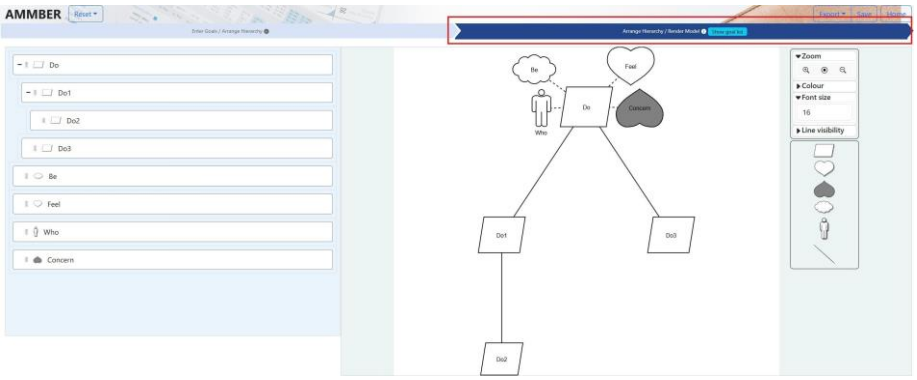


3.4 Exporting the model

Note

You can only export a model after it has been rendered. Empty models cannot be exported.

(1) Create and render your model.
(2) Go to the "Arrange Hierarchy / Render Model" page.



(3) Click  in the header and choose your desired format (PNG or SVG).

(4) A dialogue box will appear for you to select a save location on your local machine.
(5) The file will now be saved to the location you selected.

3.5 Saving the model

(1) Create your model.

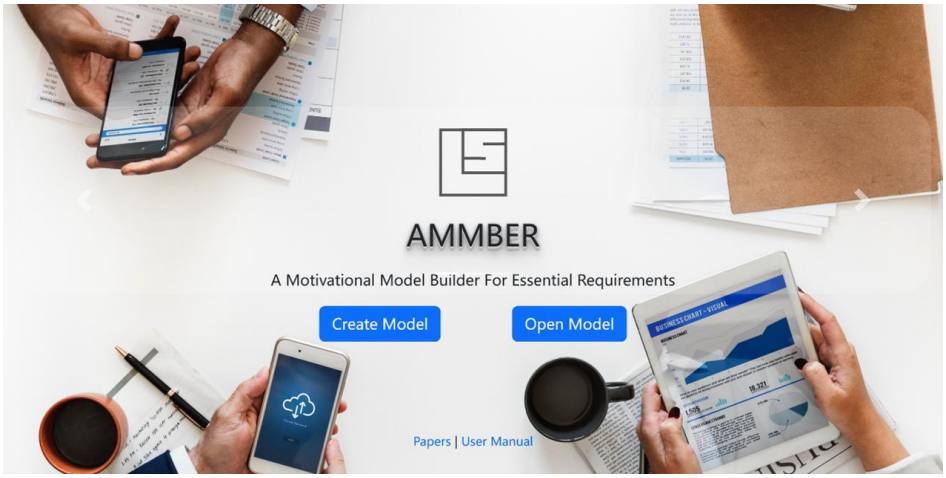
(2) in the editor page, click  in the header and choose your desired format.

(3) A dialogue box will appear for you to select a save location on your local machine.
(4) A JSON file will now be saved to your selected location for future use.

3.6 Opening a saved model

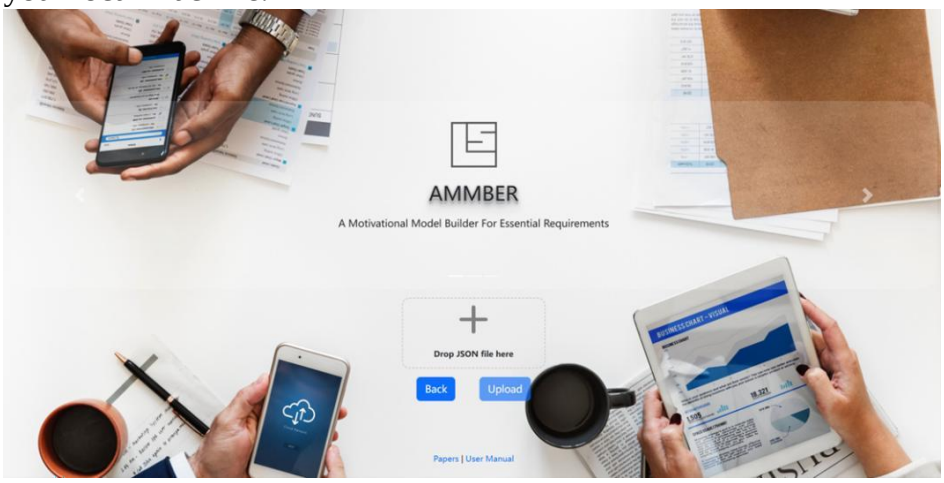
(1) You should have saved a model made by AMMBER on your local machine first. Follow this guide if you don't know how to save the model.

(2) Go to AMMBER landing (home) page.



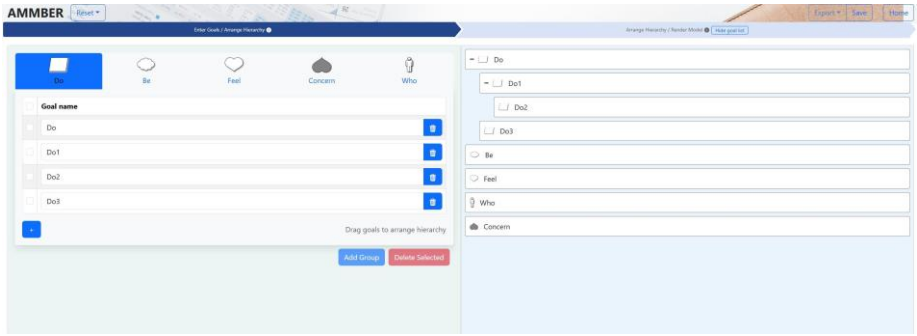
(3) Click Open Model.

(4) Click "Drop JSON file here" to open a dialogue box to select the saved model in your local machine.



(5) After selecting the saved model, the upload button will be enabled. Click Upload to upload it.

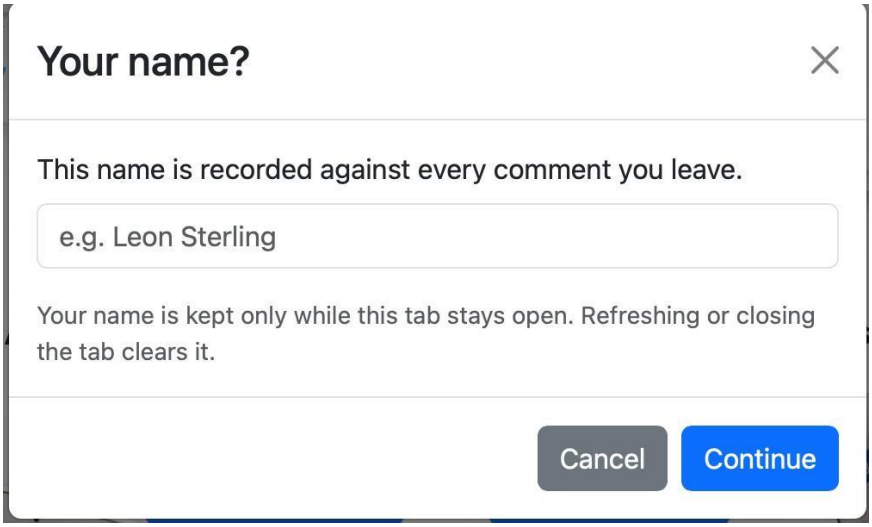
(6) You will be navigated to the "Enter Goals / Arrange Hierarchy" page with your saved model.



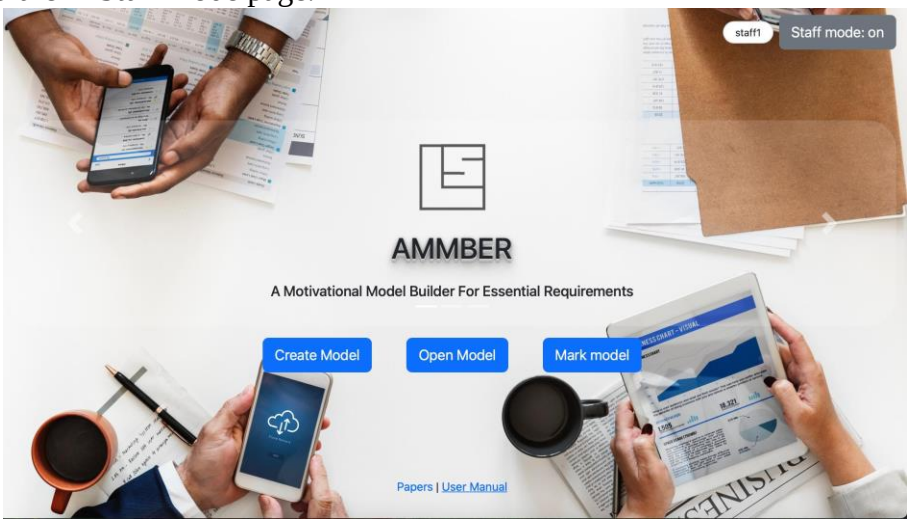
3.7 Staff Mode Instruction

(1) Click  in the top-right corner of the homepage.

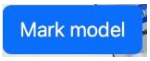
(2) Enter your name to enter Staff Mode.



(3) Now you are in Staff mode page.



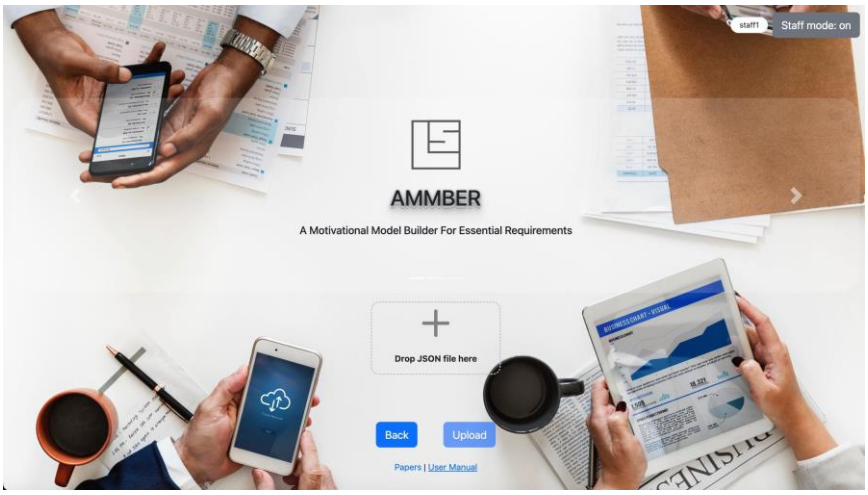
3.7.1. Adding feedback comments to model

(1) After entering the staff page, click the 

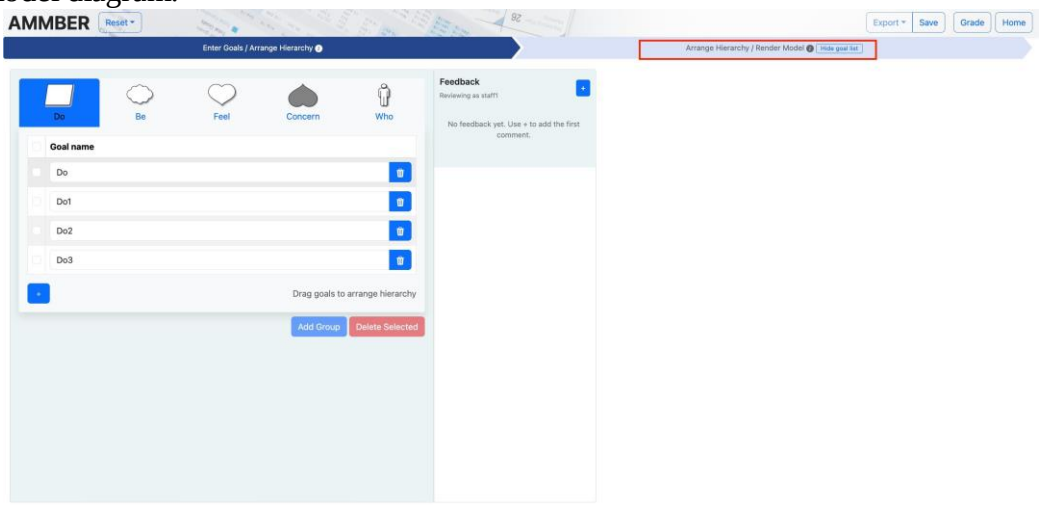
icon.

(2) Upload or drag and drop the JSON file into the drop zone, then click the 

icon to confirm.

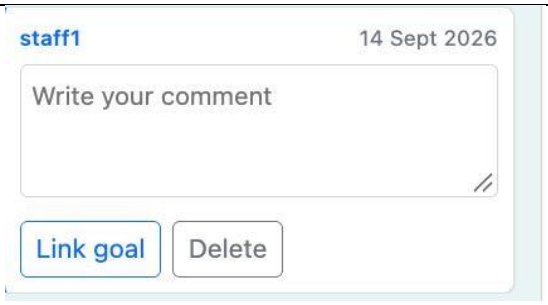


(3) Once in the feedback panel interface, click "Arrange Hierarchy / Render Model" to view the complete model diagram.

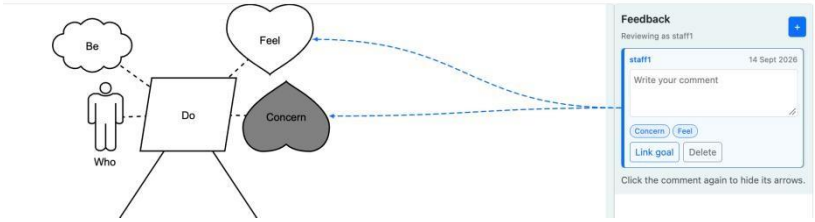


(4) Click the 
icon under "Feedback" to create a new feedback entry (multiple entries can be created at once).

(5) Enter your feedback in the text box.

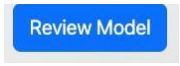


(6) Click the Link goal icon to link the feedback to the corresponding goal.

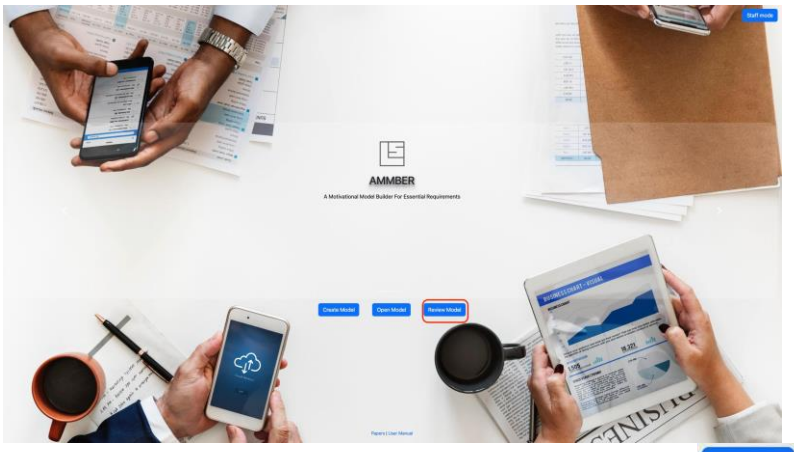


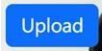
(7) Click the Delete icon to remove the feedback.

3.8 Student Reviewing Grade and Feedback

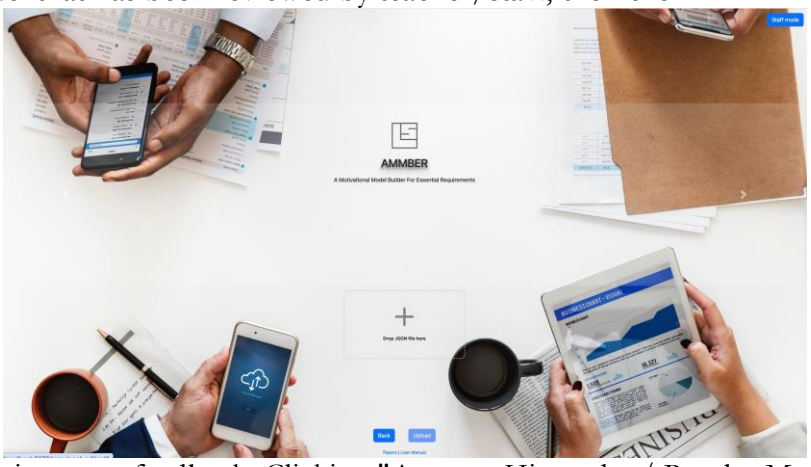
(1) Click 

in the homepage.

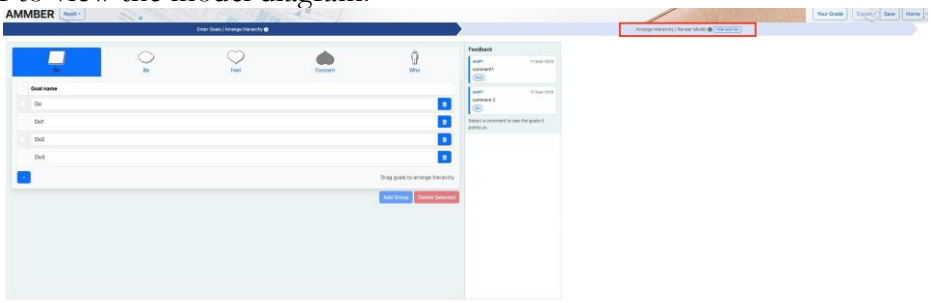


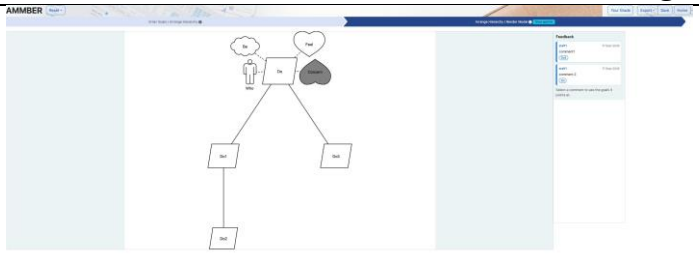
(2) Upload the model that has been reviewed by teacher/staff, then click 

to confirm.

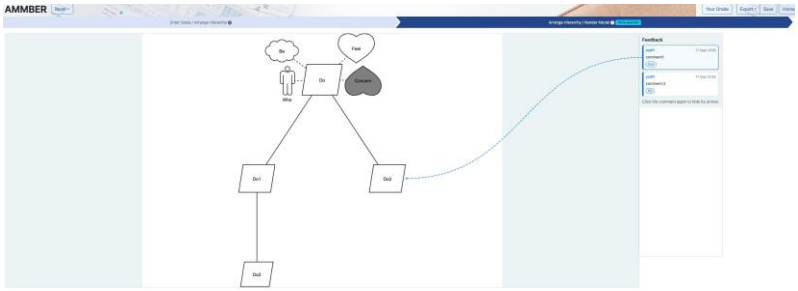


(3) Now you can review your feedback .Clicking "Arrange Hierarchy / Render Model" in the navigation bar to view the model diagram.





(4) Click a feedback comment in the Feedback panel. A dashed arrow will point from the comment to the goal(s) it refers to.

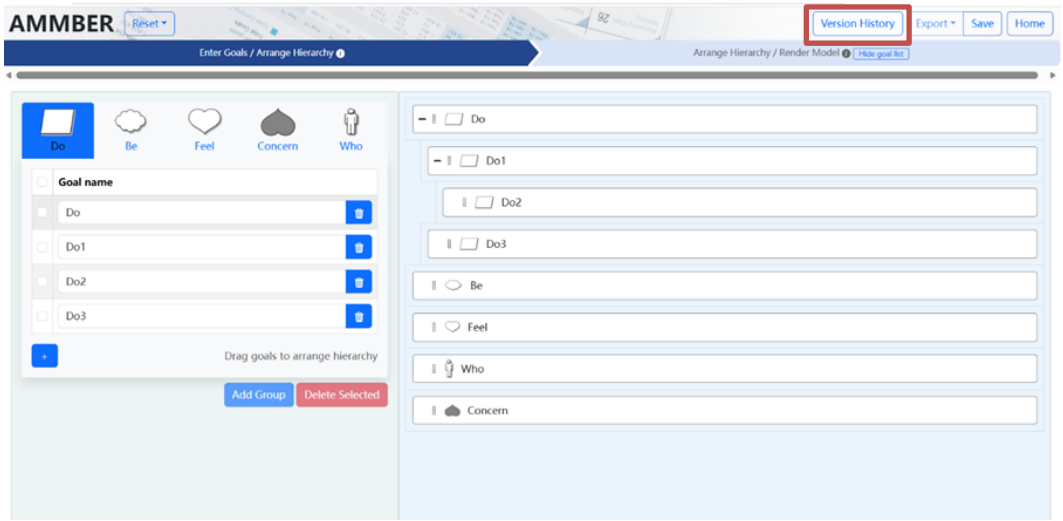


(5) Click the same comment again to hide the arrow.

3.9 Model Version Control

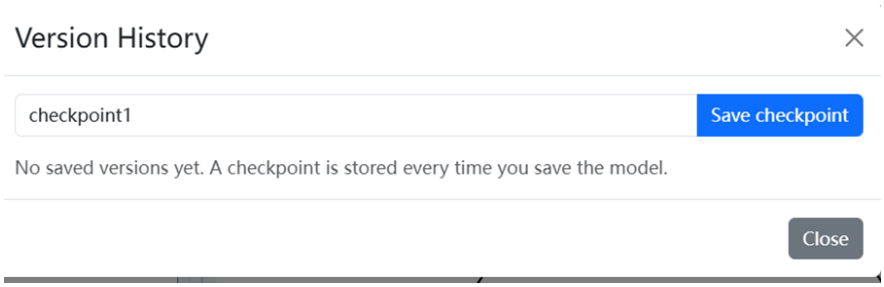
AMMBER saves historical versions (checkpoints) of each model. If a change goes wrong, or you simply want to return to an earlier state, you can restore a previous version at any time.

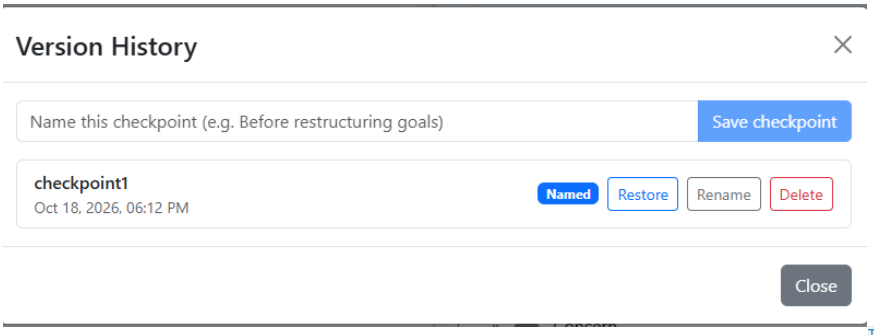
Version control is available on the "Editor – Enter Goals / Arrange Hierarchy" page via all functions.



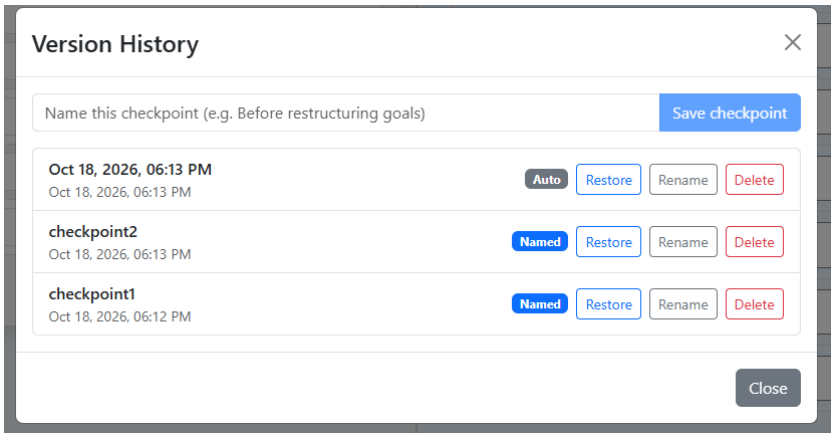
3.9.1. Saving the current model version

(1) To save the current version of your model manually, click the Version History button in the header. In the dialog that opens, type a name for the checkpoint in the text box (for example, "Before restructuring goals"), then click **Save checkpoint**. The new checkpoint appears in the list below with a **Named** badge, which marks it as a manually saved version.





(2) An automatic checkpoint is also recorded whenever a student or staff member clicks Save button to generate the JSON file, for example after a student finishes their model or a staff member finishes giving feedback. These checkpoints appear in the list with an **Auto** badge and are labelled with the date and time they were saved.



The list is sorted from newest to oldest. Auto checkpoints are created only when a file is saved. Exporting the model or opening a file does not create one.

(3). Click Restore to replace the current model with the selected checkpoint. Click Rename to change the

name of a checkpoint. Click Delete to permanently remove a checkpoint from the list.

Note

Restoring a checkpoint replaces your current model, so any changes made since your last save will be lost. If you want to keep your current work, save a checkpoint before restoring.

Note

Checkpoints are stored in your browser on this device only. They are not included in the saved JSON file, and they will not be available on another computer or browser, or after the browser data is cleared. To keep a model permanently, save it as a JSON file.

Note

AMMBER keeps up to 50 checkpoints. When the limit is reached, the oldest Auto checkpoints are removed first, and Named checkpoints are kept as long as possible.

Bibliography

[Sterling and Taveter(2009)] Leon S. Sterling and Kuldar Taveter. *The Art of Agent-Oriented Modeling*. The MIT Press, 07 2009. ISBN 9780262259217. doi: 10.7551/mitpress/7682.001.0001. URL https://doi.org/10.7551/mitpress/7682.001.0001.

[Marshall(2018)] James Marshall. Agent-based modelling of emotional goals in digital media design projects. In *Innovative Methods, User-Friendly Tools, Coding, and Design Approaches in People-Oriented Programming*, pages 262–284. IGI Global, 2018.

[Lopez Lorca et al.(2018)Lopez Lorca, Burrows, and Sterling] Antonio Lopez Lorca, Rachel Burrows, and Leon Sterling. Teaching motivational models in agile requirements engineering. In *2018 IEEE 8th International Workshop on Requirements Engineering Education and Training (REET)*, pages 30–39, 2018. doi: 10.1109/REET.2018.00010.

[L.(2017)] Sterling L. Motivational goal models – the big picture. https://static1.squarespace.com/static/61136f0c2dddc2734b652409/t/66a9c295d2a1e377d82b0c3c/1722401441685/2+Motivational+Goal+Models+%E2%80%93+The+Big+Picture.pdf, 2017.